D09. C++ Utility Pack

Overview

The C++ Utility Pack contains the standalone C++ component of the project. It supports a meaningful warehouse rule by checking high-quantity pallet placement.

The component reads runtime CSV evidence produced by the Bash layer. It does not connect directly to Oracle. This keeps the C++ part simple and easy to run locally.

Warehouse Rule

The rule is that high-quantity pallets should be assigned to active priority locations.

- **High quantity threshold: 12.**
- **Minimum location priority: 2.**
- **Required location state: active location.**

Input Files

- **runtime/snapshots/pallet_state.csv provides pallet ids, quantities, assigned locations, and final status.**
- **runtime/snapshots/location_usage.csv provides location active flags and priorities.**

Output Files

- **runtime/reports/cpp_priority_report.txt gives the summary pass/fail result.**
- **runtime/reports/cpp_priority_decisions.csv shows the detailed decision for each high-quantity pallet.**

Run Instructions

cd project
bash scripts/setup_demo.sh
bash scripts/run_demo.sh
bash scripts/run_cpp_priority_checker.sh

Expected Result

The current FlowCore scenario contains two high-quantity pallets: PLT-1001 and PLT-1003. Both are assigned to active locations with enough priority, so the expected C++ result is a pass.

FlowCore C++ Priority Placement Decision Checker
High quantity threshold: 12
Minimum location priority: 2
Priority pallets checked: 2
Result: PASS

Decision Output

pallet_id,quantity,assigned_location,location_priority,decision,message
PLT-1001,12,A01,3,APPROVED,High quantity pallet assigned to priority active location
PLT-1003,15,A03,2,APPROVED,High quantity pallet assigned to priority active location